What Separates Top-10 Picks? Public Pre-Draft Information and the NBA Draft, 2008–2025

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October 2026. Designed and directed by the author; built with AI coding tools (Claude Code). Revised October 22, 2026, after an independent data validation. Every number below is produced by the project's code and listed in its `results/key_numbers.csv`.

Abstract

Among the 1,073 players drafted from 2008 to 2025, a logistic regression on public pre-draft information (age, college production, recruiting rank, combine size, competition strength, pathway and position) separates top-10 picks from picks 11–60 with an AUC of 0.865 (95% CI 0.83–0.89) when every draft is predicted by a model fit on the other drafts. Age alone scores 0.777; the gain is +0.09 (95% CI +0.07 to +0.11). Age carries the most information: 39.4% of draftees aged 19.5 or younger went in the top 10, against 2.6% of those older than 22.5. On the 593 players with an archived post-combine mock draft (2016–2025), the mock alone scores 0.967, and adding the stats model does not measurably improve it (−0.003, 95% CI −0.011 to +0.005). The model separates players with no Division I season less well than players with one (0.758 vs 0.883, an exploratory finding). On the 2026 draft, a board saved on draft day caught 8 of the 10 top-10 picks; the consensus of three mock drafts caught 9. One draft cannot separate the two.

1. Question

Which public information available by draft night separates the players taken in the top 10 from the rest of the draft, and does any of it add to what the mock drafts already know? The study predicts *where teams draft*, not which players become good NBA players. Its sample is drafted players only, so it says nothing about who gets drafted at all.

2. Data

The unit is one drafted player. The outcome is whether he was taken with picks 1–10: 180 of 1,073 players (16.8%). Two rows in the original data were forfeited picks (2024 #59 and 2025 #60) and are removed.

Input	Source	Coverage
Draft results and birth dates	Basketball-Reference (draft pages; data archive); Wikidata where Basketball-Reference has no date	1,073 picks; birth dates 963 + 6 reviewed + 103 + 1 missing
College production and team strength	Bart Torvik player seasons and team ratings, joined by player identity with school and season checks	860 players with a D-I season
Recruiting rank	247Sports composite	613 values (505 ranked, 108 floor)
Combine size	NBA Draft Combine pages (archived)	length 504 (47%); height in shoes 351 (33%)
Mock draft	Tankathon, one archived capture per draft, May 22–June 5	10 drafts, 2016–2025

The 2020 and 2021 mock captures predate that year's combine. Drafted players missing from a mock (98) are coded pick 75 with a flag; no actual top-10 pick was missing from any mock. Two inputs (age versus draft class and the shooting composite) are scaled within each draft's pool of draftees, which is known on draft night.

3. Methods

- **Models.** A regularized logistic regression is the headline model; gradient boosting is fit alongside it. Each draft is scored by models fit on the other 17 drafts (leave one draft out).
- **Baselines.** A logistic regression on age alone, and the post-combine mock pick as published, with no fitting.
- **Metrics.** AUC: the chance that a randomly chosen top-10 pick is ranked above a randomly chosen pick 11–60 (0.5 is a coin flip). Brier score for probabilities (lower is better).
- **Uncertainty.** Intervals resample whole drafts (2,000 bootstrap draws), because players in one draft share a pool and a mock. Comparisons between models are paired differences on the same players. Counts of 10 get Wilson intervals.
- **Declared in advance.** The robustness rows and the subgroup screen were written down before they were run (`docs/analysis_choices.md`). Choices made after seeing the data are listed there too.

4. Results

4.1 The stats model against age alone

Model (all 1,073 players)	AUC	95% CI	Brier
Age alone (fitted)	0.775		
Stats model, logistic (headline)	0.865	0.83–0.89	0.097
Stats model, gradient boosting	0.844		
Average of the two	0.864		
Constant guess	0.500		0.140

Gain of the logistic stats model over the fitted age model: +0.090 AUC (95% CI +0.069 to +0.110). Raw age alone, unfitted, scores 0.777 on the 1,072 players with a birth date.

Across the 18 drafts, 100 of the 180 top-10 picks land in the model's top 10 for their draft (0.556 per draft), against 71 for age alone.

4.2 Age on draft night

Younger draftees are taken in the top 10 far more often (Figure 2): 39.4% (69 of 175) of those 19.5 or younger, 30.2% at 19.5–20.5, 8.3% at 20.5–21.5, 5.2% at 21.5–22.5 and 2.6% (5 of 191) older than 22.5. A logistic fit on exact age falls steadily; among draftees, a year of age goes with about 76% lower odds of a top-10 pick, about 13 points of probability on average. This is an association among players teams chose to draft, not the effect of being younger. The 20.5–21.5 share depends on where its cut points fall, so the continuous curve is the better summary.

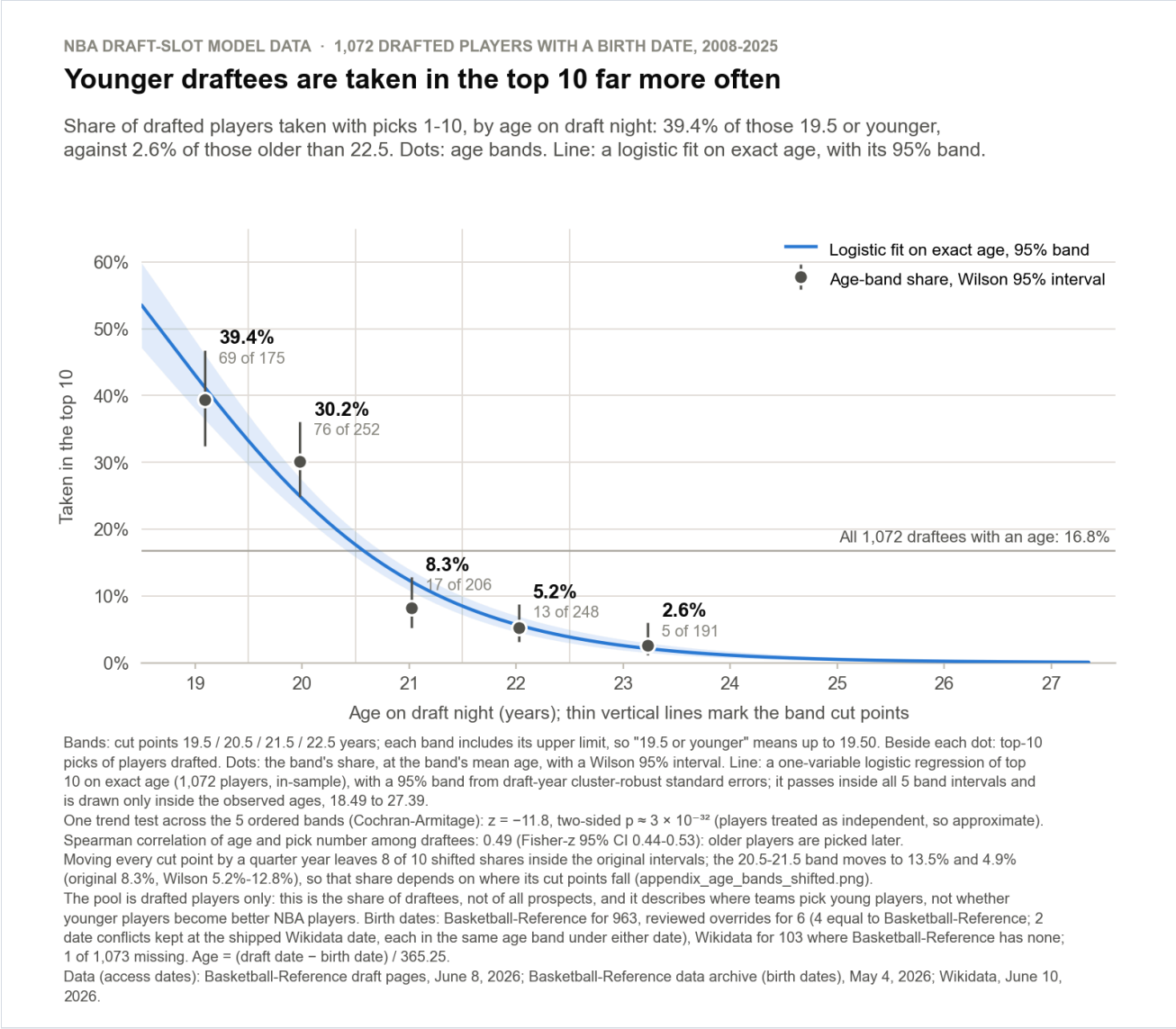


Figure 2. Share of drafted players taken with picks 1–10 by age on draft night (Wilson 95% intervals), with a logistic fit on exact age and its 95% band. 1,072 drafted players with a birth date, 2008–2025.

4.3 Against the post-combine mock draft

The mock draft is the hardest baseline. On the same 593 players (2016–2025, 100 top-10 picks), the leave-one-draft-out AUCs are 0.761 for age alone, 0.846 for the stats model, 0.967 for the mock as published and 0.964 for the stats model with the mock pick added (Figure 1, panel A). The paired difference between stats-plus-mock and the mock alone is −0.003 (95% CI −0.011 to +0.005): the stats model does not measurably improve the mock. The design would detect a gain of 0.011 about 80% of the time. On probabilities, the mock-only model's Brier score is 0.0501 against 0.0537 for stats-plus-mock, a difference of +0.0036 (95% CI −0.0016 to +0.0087), also within noise. Across 2016–2025 the stats-plus-mock model caught 81 of the 100 top-10 picks and the mock 82.

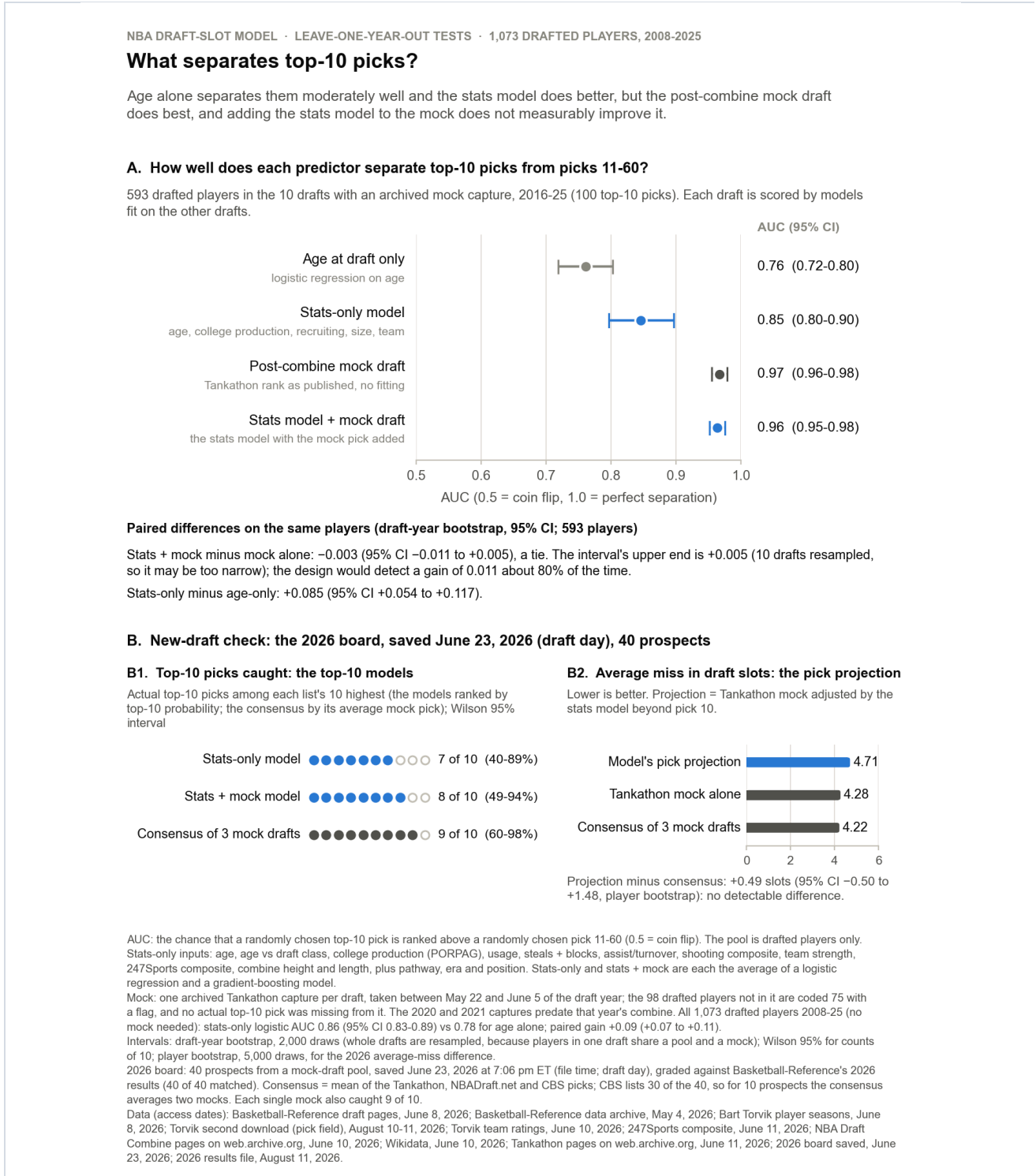


Figure 1. Panel A: leave-one-draft-out AUC on the same 593 players with a mock capture (draft-year bootstrap 95% intervals). Panel B: the 2026 board saved on draft day, graded against the actual draft.

4.4 Where the model is weaker (exploratory)

A screen of nine subgroups, declared before it was run, found one detectable difference: the model separates players with no Division I season less well (AUC 0.758, n = 213) than players with

one (0.883, n = 860), a difference of +0.125 (95% CI +0.043 to +0.221). It has no college statistics for them. Guards (0.859), bigs (0.869) and wings (0.896) are not detectably different (wing minus guard +0.037, 95% CI −0.029 to +0.103), and neither are the eras (2008–13 minus 2020–25: +0.046, −0.019 to +0.108). One finding out of nine screened can be chance, so this is a lead, not a result.

4.5 What carries the prediction

Out-of-fold permutation importance (the drop in AUC when an input is shuffled within the held-out draft) is largest for the two age inputs together (0.269), then the college production block (0.107). Combine size adds nothing measurable: dropping it changes AUC by +0.002 (95% CI −0.002 to +0.007) on all players, and by the same amount on the 504 players with a real measurement. Importance here is predictive weight among draftees, not an effect, and noisy inputs such as one season of college three-point shooting are understated.

4.6 One new draft: 2026

A board of 40 prospects was saved on draft day (June 23, 2026) and graded against the actual draft. The stats-only model caught 7 of the 10 top-10 picks (Wilson 95% interval 40%–89%), the stats-plus-mock model 8 (49%–94%) and the consensus of three mock drafts 9 (60%–98%). For the pick projection, the average miss was 4.71 draft slots, against 4.28 for Tankathon alone and 4.22 for the consensus; the projection-minus-consensus difference is +0.49 slots (95% CI −0.50 to +1.48). One draft cannot separate these. Eighteen of the 40 prospects sit at least partly outside what the model was fit on (no D-I season, an imputed recruiting or combine value, or an input outside the 2008–25 range).

5. Checks and diagnostics

- **Walk-forward.** Scoring each draft only with models fit on earlier drafts gives 0.847 against 0.858 for leave-one-draft-out on the same 2012–25 players. Part of that gap is the smaller training set, so leave-one-draft-out is at most slightly optimistic.
- **Calibration.** The probabilities rank players well but do not sum to 10 in every draft: per-draft sums run 6.30 to 12.31. Rescaling each draft to 10 lowers the Brier score from 0.0971 to 0.0961 (Appendix, Figure A1). The ranking within each draft is unaffected.
- **Few drafts.** With only 10 mock-era drafts, bootstrap intervals can be too narrow. A post-hoc leave-one-draft-out jackknife gives −0.013 to +0.006 for the increment over the mock, and no single left-out draft moves it above zero.
- **Codings.** Other codings of players missing from the mock (log pick plus a flag; missing pick plus a flag) give the same tie.
- **Second route.** The headline AUCs are recomputed a second way (Mann–Whitney U), and Brier scores and Wilson intervals by hand, on every run.

6. Data corrections

An independent validation on October 21–22, 2026 checked every dataset against outside sources. The draft results matched two independent sources pick for pick. The college join did not: a name-matching fallback gave 30 drafted players another person's college season, 12 of them top-10 picks, and 5 birth dates belonged to namesakes. The revision joins players by identity with school and season checks, takes birth dates from Basketball-Reference first, and removes the two forfeited-pick rows. The effects:

Quantity	Before	After
Drafted players	1,075	1,073
Stats model AUC (logistic)	0.848	0.865
Guards' AUC	0.790	0.859
Mock-era players / mock AUC	595 / 0.968	593 / 0.967
Stats + mock minus mock	−0.001	−0.003

The earlier claim that guards are hardest to sort came from the wrong college rows; it does not survive the fix.

7. Limitations

Limit	Mechanism	Likely direction
Drafted players only	The sample is a pool teams already selected.	Says nothing about who gets drafted; associations can differ among all prospects.
No college inputs without a D-I season	Their college inputs are missing by construction.	Weaker separation for internationals and G League or prep players.
Box scores under-measure defense	College box scores miss most defensive value and fit.	Hypothesis: mainly defensive players are ranked too low.
Noisy proxies	Recruiting ranks and one season of shooting are measured with error.	Their weight is understated.
Early mocks in 2020–21	Those captures predate the combine.	Weakens the mock baseline, which can only flatter the model; the tie holds.
Few drafts	Only 18 drafts (10 with a mock) to resample.	Stated intervals are somewhat too narrow.
Information teams have	Interviews, medicals and workouts shape the draft.	The model can only reach the part public data explain.

8. How a team could use it

As a public-data prior for where a prospect is likely to be drafted, checked against the board. Where public statistics and the mock draft disagree, that is a question for scouts, not an answer. The model adds nothing once the mock is known, and it is weakest for players without a college record, which is where scouting matters most.

9. Reproducibility and how this was built

One command, `python run_all.py`, rebuilds every result and figure from cached inputs in about five minutes without network access. It checks the inputs' fingerprints, re-derives the headline numbers a second way and runs the test suite (144 tests pass). The raw third-party pulls and the frozen 2026 draft-day files stay in a private data folder and are not shared.

The author chose the question and the tests each result had to pass, and directed AI coding tools (Claude Code) that wrote most of the code and drafted this paper from the project's results files. The revision applied lessons from Syracuse coursework in sport data analysis, statistics, database design and communication: identity keys instead of names for joins, resampling whole drafts, intervals on differences, declared robustness checks, and reporting nulls as bounds. Every change is logged with its evidence in the repository.

Data sources (access dates)

- Basketball-Reference draft pages, June 8, 2026; Basketball-Reference data archive (birth dates), May 4, 2026.
- Bart Torvik player seasons, June 8 and August 10–11, 2026; team ratings, June 10, 2026.
- 247Sports composite, June 11, 2026. NBA Draft Combine pages via web.archive.org, June 10, 2026. Wikidata, June 10, 2026.
- Tankathon mock drafts via web.archive.org, June 11, 2026. 2026 draft-day board saved June 23, 2026; 2026 results file, August 11, 2026.

Appendix

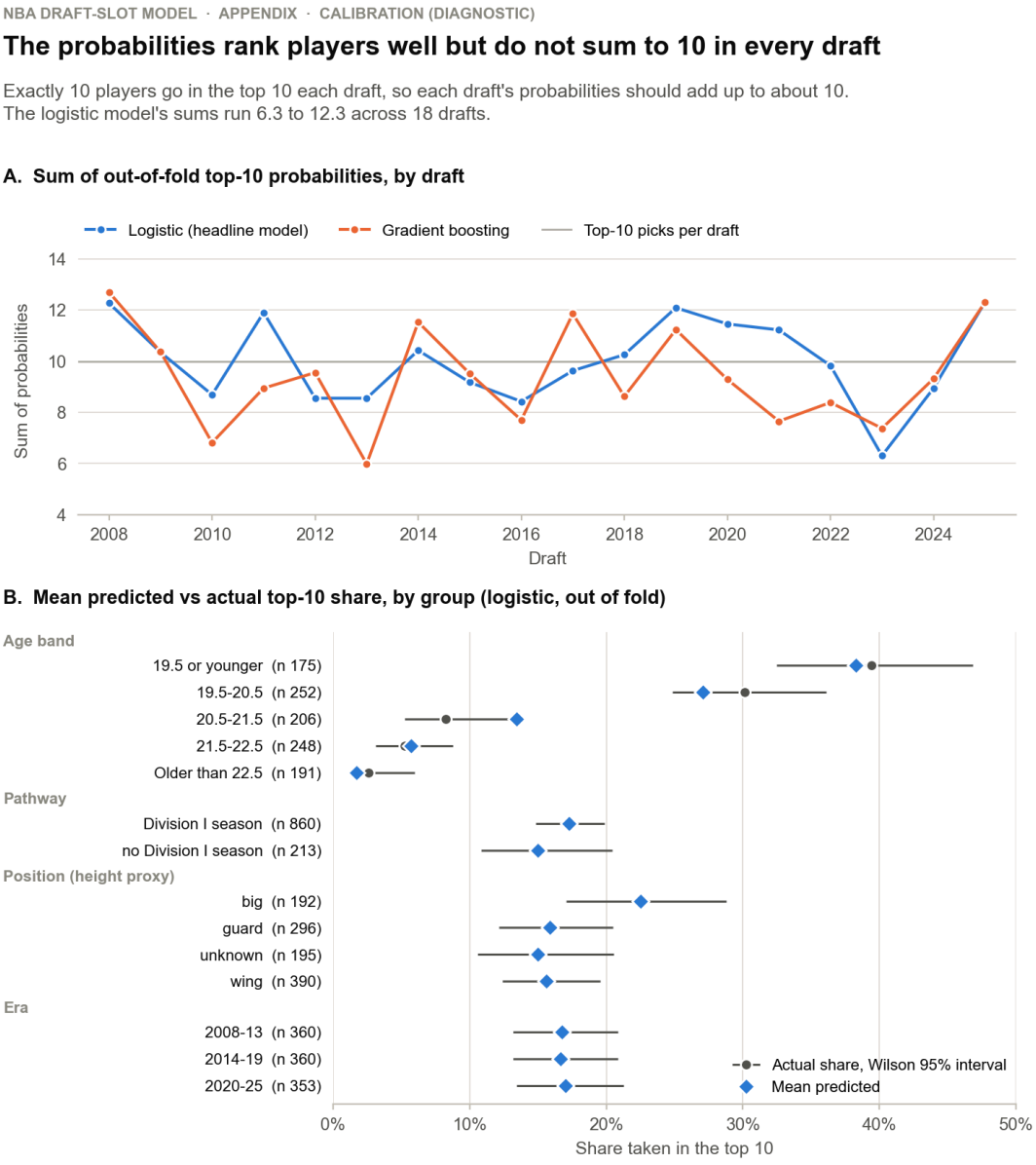


Figure A1. Calibration diagnostic. Panel A: each draft's out-of-fold top-10 probabilities, summed (every draft has exactly 10 top-10 picks). Panel B: mean predicted against actual top-10 share by group, with Wilson 95% intervals.